

# Technical Report:

## Part1: Test Setup

**Please explain what are the aspects that one would need to consider before starting such a test. Please make sure to include test split, test duration and measurement criteria.**

- Crucial steps before starting the tests are
  - Ensure alignment with what the objective/goal, null hypothesis, alternate hypothesis, and what Conversion means: Here it is 'Signup conversion rates'.
  - **Test split:** Randomly split between the two groups 50-50% ( A/B testing) or into three groups (34/33/33% split) for A/B/C testing with random(randomization) assignment to avoid bias and ensure fairness. Use pre-test power calculations to determine required sample size for each group factoring expected base CR and minimum detectable effect(MDE).
  - **Test duration:** Run the test for a minimum of 2-3 full business cycles (at least 2 weeks including weekday and weekends behaviours). The duration should also meet the required sample size to detect a meaningful difference with high confidence. Lesser durations may make the test unreliable while running too long introduces external noises and seasonality.
  - **Measurement criteria:** defining primary and secondary KPI's with success metrics and ensuring proper tracking implementation infrastructure. Also past data to compare shifts.
- **What would you need to consider to ensure that the split avoids bias?**
  - **Pre-Test A/A checks:** Running an A/A test to validate that metrics don't differ significantly before the variant changes are introduced.
  - **Randomization:** Assign users randomly (not based on location , device, or time) and make sure all segments (countries/devices) have near uniform distribution.
  - Assignment should happen ideally at user level. This ensures the returning user consistently sees the same variant throughout the test duration.
  - Run the tests long enough to avoid seasonality bias to account for daily/weekly trends.
  - Avoid overlapping with other test or experiment periods.
  - Exclude internal traffic.
- **How can one ensure that we run the test for a sufficient period to capture variations in user behaviour?**
  - The primary driver of duration is reaching the required sample size per variant. And this is calculated based on statistical power analysis
  - Once we know the required sample size, estimate how long it will take to reach the number of unique visitors/sessions per variant based on historical LP traffic. Ensure the test runs for at least 2 full business cycles (2-3 weeks) to capture variations in weekdays and weekends.
  - While sample size determines statistical significance, running for a longer, representative period increases confidence that the observed effect isn't just a temporary anomaly.
  - Avoid stopping tests early due to premature analysis.
- **What would be appropriate primary and secondary KPIs to measure the success of the test?**
  - The primary KPI is the Conversion rate  
Conversion Rate = (Number of sessions with at least 1 Signup/Total sessions) \* 100
  - Secondary KPI's (Depending on the data available)
    - \* Paid Conversion Rate = (Number of sessions with at least 1 paid subscription/Total session) \* 100
    - \* Click Through Rate - How effective it drives to next step
    - \* Paid to click Ratio - How effectively users are converting to paid
    - \* Average order value - Average value of

- \* Bounce Rate
- \* Average session duration

**What type of data one would need to collect to be able to analyse the results?**

| User/Session level | Engagement/<br>Behaviour level | Conversion level | Demographic level   | User Acquisition level |
|--------------------|--------------------------------|------------------|---------------------|------------------------|
| Session data       | Clicks Data                    | Signups/Action   | Device type         | Traffic Source         |
| Timestamp/Dates    | Views                          | Purchases        | Geographic location | Returning vs New User  |
| Variant Types      | Time on page                   |                  |                     |                        |

## Part2: Test Analysis

### Data Analysis

- **Have a first quick look at the data to familiarise yourself with it and identify if there are any quality issues or anomalies.**
- After thoroughly reviewing the dataset, which contains 618,875 rows, 12 columns of data with Excel and Python (Notebook).
- several key quality issues and data anomalies were identified.
  - Missing values: 277 sessions of missing or \* values in the Country column. Also there is a country xxx which is not known. These sessions cannot be used reliably for country based conversion analysis.
  - Duplicate SessionID's: There are 29,821 duplicate sessionID's. Since sessionID represents a session/visit, repeated sessions could have bias and make the data skewed and this could probe an issue in our analysis. Hence the data must be deduplicated (keeping highest signups or revenue and removing other duplicates based on sessionID).
  - Outliers: sessionDurationInSeconds has an outlier with a value of 56,328 seconds (15+ hours). This is probably an idle session.
  - Also the column index is of no use as it just represents the index of rows. This should be dropped.
  - The column 'subscriptionsTotalByCycle' contains values in {} dictionary format. Cleaning this to just store the value inside and keeping 'NaN' if null.
  - There are sessions with multiple (> 1) 'sign ups' and multiple 'subscriptionsTotal' per session. Since these are session based data, all conversion rates (signup and paid) are calculated using binary session level. i.e.,  
A session is marked 'a signup' if signups > 0  
A session is marked 'a paid conversion' if subscriptionsTotal > 0
- **Given the hypothesis above and the data provided, what KPI would you look into? What data may be missing that would be important for this task?**
- **Primary KPI:**  
**signup conversion rate (CR).** The percentage of sessions that result in signups as this is aligned with the hypothesis goal.  
Signup Conversion Rate = (Number of sessions with at least 1 Signup/Total sessions) \* 100
- **Secondary KPI's:**
  - \* Paid Conversion Rate. The percentage of sessions that led to a paid subscription.  
Paid Signup Rate = (Number of sessions with at least 1 paid subscription/Total session) \* 100
  - \* **Subscriptions-to-Sign Up Ratio (%):** The proportion of signups that converted into paid subscriptions, indicating signup quality.  
Subscriptions-to-Sign Up Ratio (%) = (Number of sessions with at least 1 paid subscription/ Number of sessions

with at least 1 Signup) \* 100

\* Average session duration (sec): The average amount of time users spent on LP, per session, reflecting engagement.

\* Bounce Rate (%) = The percentage of sessions with minimal interaction.

i.e., Bounce being defined as a session with 0 seconds or  $\leq 1$  page views with no signups and purchases.

- However, **some data is missing** that would strengthen the analysis, including:
  - **Traffic source** (paid, direct, social etc),
  - **Returning vs New User flag** (to track repeat session behavior)
  - Engagement Metrics (Clicks) to find click through rates
- **Moving forward, the primary KPI is Conversion Rate, which is the percentage of visitors who sign up. Calculate the Conversion rate per Country. What is the country that performs the best?**

The conversion rate per Country is calculated and the all the conversion rates per country data table is attached in the [Appendix for reference](#), segregated by countries with minimum 1000 sessions and countries with  $>100$  and  $< 1000$  sessions to segregate the high volume and high impact countries and the countries with  $< 100$  sessions is excluded since the data sample is not significant to provide analysis with confidence.

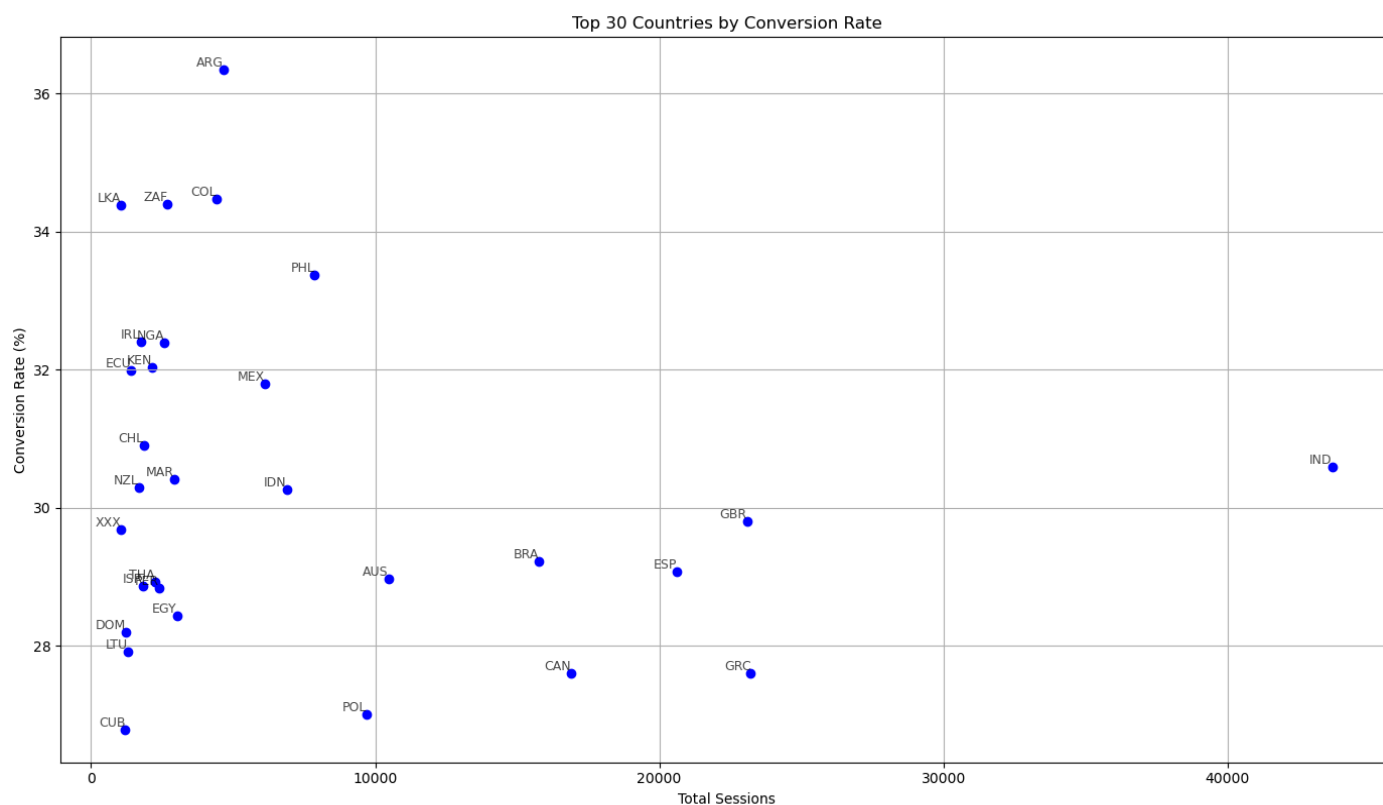
For countries with minimum **1000 sessions**,

**Argentina (ARG)** has the highest signup conversion rate of **36.34%** with 4,675 total sessions.

For countries with range between **100 to 1000 sessions**,

**Maldives (MDV)** has the highest Signup Conversion rate of **37.82%** with 119 total sessions.

#### Plot of Top 30 Country -Wise Signup Conversion Rates with minimum 1000 sessions



- **What is the overall conversion rate?**

- The overall conversion rate is **26.11%** (including all sessions, even those without country info)

| total_sessions | signedup_sessions | conversion_rate_% |
|----------------|-------------------|-------------------|
| 589,054        | 153,773           | 26.11             |

- The **overall conversion rate per variant** (including all sessions, even those without country info):

| Variant | total_sessions | signedup_sessions | conversion_rate_% |
|---------|----------------|-------------------|-------------------|
| A       | 194,446        | 50,873            | 26.16             |
| B       | 194,025        | 50,258            | 25.9              |
| C       | 200,583        | 52,642            | 26.24             |

Hypothesis:

**Null Hypothesis (H<sub>0</sub>):** There is no difference in signup conversion rates between Variant A (control) and the test variants B and C.

**Alternative Hypothesis (H<sub>1</sub>):** At least one variant has a significantly different signup conversion rate compared to control A.

Since there is small difference in Signup Conversion Rates between A vs C, validating it with with 2 proportion Z test statistical tests to test statistical significance

Using Statsmodel library in Python,

| Comparis on | CR A (%) | CR Variant (%) | Uplift (%) | Z-Stat | p-value | Significant?  |
|-------------|----------|----------------|------------|--------|---------|---------------|
| A vs B      | 26.16    | 25.9           | -0.26%     | 1.85   | 0.0646  | No (p > 0.05) |
| A vs C      | 26.16    | 26.24          | 0.08%      | -0.58  | 0.5606  | No (p > 0.05) |

**Neither B nor C shows a statistically significant difference** from control A. Null hypothesis is not rejected for primary KPI.

Even though **Variant C has the highest conversion rate**, the **p-value of 0.56** suggests this difference could easily be due to chance.

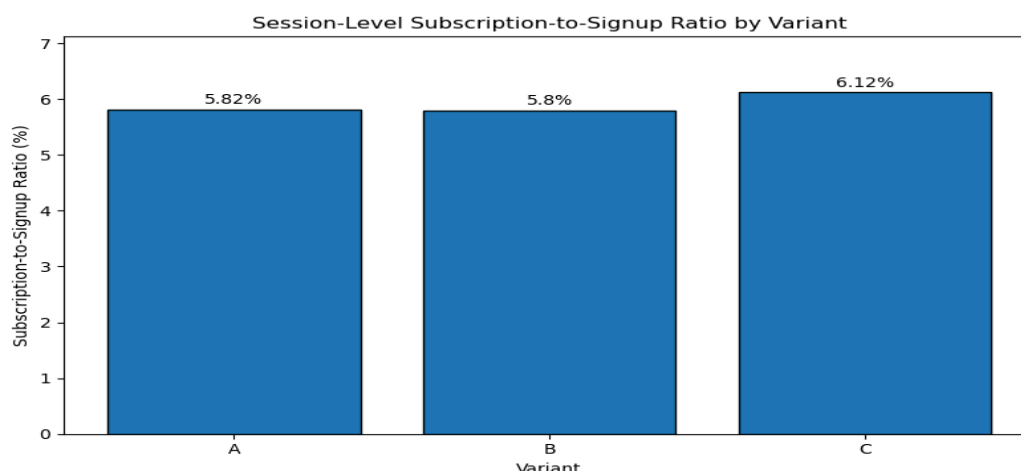
#### **Paid Conversion Rate per Variant**

| flag | total_sessions | paid_converted_sessions | paid_conversion_rate_pct |
|------|----------------|-------------------------|--------------------------|
| A    | 194,446        | 4,044                   | 2.08                     |
| B    | 194,025        | 3,993                   | 2.06                     |
| C    | 200,583        | 4,358                   | 2.17                     |

- Validating the result with Z tests to test statistical significance. The test result is in the appendix.  
The **Variant C outperforms Control A** in terms of **paid conversions** (despite a small uplift of +0.09%) with  $p = 0.043 (< 0.05)$ .
- If the goal is **maximizing paid conversions**, Variant **C should be prioritized** over A or B — even if the signup CR difference wasn't statistically significant.

## Subscription to Sign Up Ratio (session based) (% of signups resulted in paid subscription)

### Bar Plot of Subscription to sign up ration across variants



**Variant C** again performs best converting **6.12% of signup sessions into paying subscribers (+0.30% uplift)**. This confirms that not only does **Variant C** drive more paid conversions, but also does so **more efficiently** per signup.

### What other learnings can you highlight from the data?

**Other Learnings:** (The data table for each of these analysis is added in the appendix)

- Performing Country level split of Conversion Rates of variants, control group and measuring the delta. Regions like **UAE, Chile, New Zealand** have had better Signup Success Rates with Control Group A (delta +3%) compared to test groups.
- Regions like **Thailand (THA)** and **Malaysia(MYS)** etc responded well to Variant B (delta +4.5%) compared to other groups. These findings suggest that users showcase a low risk mindset and are experimental driven and maybe with less purchasing power.
- Conversion rates across desktops (~27%) are higher than phones (~22%) and are nearly consistent with slightly higher values to variant C groups, showcasing better visibility, more information options and easier process.
- The bounce rates across all variants is around (~6%). This suggests that plan order has minimal impact on immediate user drop-off behavior.
- Average session duration is 256 seconds (~4.27mins) and is consistent across all plan orders and not impacted.
- Majority of Users choose **1 month plan** across each variant with distribution of ~62-64%, followed by 12 month plan, and then 24 month plan. ~3% of its total orders of Variant C was 3 month plans.
- Time series trends across each variant showed no major trends and performed consistently for the duration of tests.
- Weekdays had slightly better conversions than on weekends which could be explained due to availability and higher intent.
- Countries like **Oman (4.7%), Qatar (9.2%)** have the lowest signups conversion rate. More user behavior and interaction data is needed to access and understand the motives and plan better engagement strategies. Also providing added value or benefits or promotions to encourage signups and conversions could lead to better Conversion rates

## Part3: Final recommendation

- Considering the primary KPI and other secondary KPIs of your choice, what variant would you recommend to accept? Why?

### Key KPI Comparison

| KPI                                | Variant A (Control) | Variant B         | Variant C             |
|------------------------------------|---------------------|-------------------|-----------------------|
| Signup Conversion Rate (%)         | 26.16               | 25.90             | <b>26.24</b> (+0.08%) |
| Paid Conversion Rate (%)           | 2.08                | 2.06              | <b>2.17</b> (+0.09%)  |
| Subscriptions-to-Sign Up Ratio (%) | 5.82                | 5.80              | <b>6.12</b> (+0.30%)  |
| Avg Session Duration (sec)         | 256.36 (4.27min)    | 255.18 (4.25 min) | 256.21 (4.27 min)     |
| Bounce Rate (%)                    | 6.08                | 6.24              | 6.17                  |

Although the **Signup Conversion Rate**(primary KPI) did not show a statistically significant improvement, **Variant C** performed well over **Control group A** across monetization and secondary KPI's. It attracts **higher quality users, also attracts best value for money users, leads to more paid conversions per signups and delivers the same or slightly more signups/conversions** compared to Control group A for free users while maintaining solid engagement levels.

Also country-level analysis revealed that some regions, such as **UAE, Chile, and New Zealand**, performed better with the **Control Variant (A)**, while countries like **Thailand and Malaysia** responded best to **Variant B** suggesting a **localized strategy** for better results in such regions.

### Recommendation:

I would recommend **Variant C** (24/12/1) for **primary rollout**. Although, some regions, such as **UAE, Chile, and New Zealand**, performed better with the **Control Variant (A)**, while countries like **Thailand and Malaysia** responded best to **Variant B** suggesting these variants should be continued with a **localized strategy** for better results in such regions.

## Appendix

Conversion Rates per Country (minimum 1000 sessions)

| Country | total_sessions2 | Converted_sessions2 | Conversion_rate% |
|---------|-----------------|---------------------|------------------|
| ARG     | 4,675           | 1699                | 36.34            |
| COL     | 4,410           | 1520                | 34.47            |
| ZAF     | 2,683           | 923                 | 34.4             |
| LKA     | 1,053           | 362                 | 34.38            |
| PHL     | 7,863           | 2624                | 33.37            |
| IRL     | 1,741           | 564                 | 32.4             |
| NGA     | 2,581           | 836                 | 32.39            |
| KEN     | 2,148           | 688                 | 32.03            |
| ECU     | 1,416           | 453                 | 31.99            |
| MEX     | 6,106           | 1942                | 31.8             |
| CHL     | 1,857           | 574                 | 30.91            |
| IND     | 43,689          | 13366               | 30.59            |
| MAR     | 2,910           | 885                 | 30.41            |
| NZL     | 1,680           | 509                 | 30.3             |
| IDN     | 6,888           | 2085                | 30.27            |
| GBR     | 23,091          | 6881                | 29.8             |
| BRA     | 15,747          | 4602                | 29.22            |
| ESP     | 20,615          | 5995                | 29.08            |
| AUS     | 10,461          | 3031                | 28.97            |
| THA     | 2,251           | 651                 | 28.92            |
| ISR     | 1,815           | 524                 | 28.87            |
| PER     | 2,406           | 694                 | 28.84            |
| EGY     | 3,017           | 858                 | 28.44            |
| DOM     | 1,227           | 346                 | 28.2             |
| LTU     | 1,308           | 365                 | 27.91            |
| CAN     | 16,879          | 4658                | 27.6             |
| GRC     | 23,220          | 6408                | 27.6             |
| POL     | 9,714           | 2624                | 27.01            |

|     |         |       |       |
|-----|---------|-------|-------|
| CUB | 1,202   | 322   | 26.79 |
| TWN | 2,110   | 557   | 26.4  |
| MYS | 3,805   | 1003  | 26.36 |
| ROU | 3,440   | 899   | 26.13 |
| USA | 121,701 | 31464 | 25.85 |
| SRB | 1,345   | 347   | 25.8  |
| NOR | 3,486   | 896   | 25.7  |
| BGR | 1,227   | 315   | 25.67 |

Conversion Rates per Country (between 100 and 1000 sessions)

| Country | total_sessions2 | Converted_sessions2 | Conversion_rate% |
|---------|-----------------|---------------------|------------------|
| MDV     | 119             | 45                  | 37.82            |
| ZMB     | 125             | 45                  | 36               |
| JAM     | 359             | 127                 | 35.38            |
| ZWE     | 345             | 122                 | 35.36            |
| TTO     | 241             | 83                  | 34.44            |
| BOL     | 625             | 212                 | 33.92            |
| MUS     | 160             | 54                  | 33.75            |
| PRY     | 315             | 106                 | 33.65            |
| URY     | 549             | 182                 | 33.15            |
| CRI     | 593             | 192                 | 32.38            |
| PRI     | 336             | 107                 | 31.85            |
| TUN     | 908             | 289                 | 31.83            |
| TGO     | 124             | 38                  | 30.65            |
| PAN     | 422             | 128                 | 30.33            |
| BIH     | 351             | 105                 | 29.91            |
| CMR     | 218             | 65                  | 29.82            |
| HND     | 246             | 73                  | 29.67            |
| GHA     | 730             | 213                 | 29.18            |
| MDG     | 381             | 111                 | 29.13            |



## Paid Conversion Rates Z-test Results:

| Comparison | CR A (%) | CR Variant (%) | Uplift (%) | Z-Stat | p-value | Significant?       |
|------------|----------|----------------|------------|--------|---------|--------------------|
| A vs B     | 2.08     | 2.06           | -0.02      | 0.48   | 0.6336  | No ( $p > 0.05$ )  |
| A vs C     | 2.08     | 2.17           | 0.09       | -2.02  | 0.043   | Yes ( $p < 0.05$ ) |

The test shows that Variant C uplift is statistically significant ( $p < 0.05$ )

## Conversion Rates of Variants across device type

|      | Desktop        |                   | Phone          |                   |
|------|----------------|-------------------|----------------|-------------------|
| Flag | total_sessions | conversion_rate_% | total_sessions | conversion_rate_% |
| A    | 148,580        | 27.31             | 45,866         | 22.46             |
| B    | 147,838        | 27.01             | 46,187         | 22.36             |
| C    | 153,076        | 27.35             | 47,507         | 22.7              |

## Conversion Rates of countries across Variant with large Delta

Countries where Variant A (1/24/12) performed best

| Country | total_sessions<br>2 | Converted_sessions2 | Conversion_rate% |                 |                 |
|---------|---------------------|---------------------|------------------|-----------------|-----------------|
| Country | CR_A (%)            | CR_B (%)            | CR_C (%)         | $\Delta$ A vs B | $\Delta$ A vs C |
| ARE     | 27.37               | 23.41               | 23.23            | <b>3.96</b>     | <b>4.14</b>     |
| BGD     | 26.82               | 24.56               | 24.55            | 2.26            | 2.27            |
| CHL     | 33.17               | 29.08               | 30.37            | <b>4.09</b>     | 2.8             |
| EGY     | 30.12               | 26.75               | 28.43            | 3.37            | 1.69            |
| IRL     | 33.69               | 31.36               | 32.14            | 2.33            | 1.55            |
| ISR     | 30.7                | 27.24               | 28.71            | 3.46            | 1.99            |
| NZL     | 32.35               | 27.24               | 31.4             | <b>5.11</b>     | 0.95            |

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Countries where Variant B (1/12/24) performed best

| Country | total_sessions<br>2 | Converted_sessions2 | Conversion_rate% |                 |                 |
|---------|---------------------|---------------------|------------------|-----------------|-----------------|
| Country | CR_A (%)            | CR_B (%)            | CR_C (%)         | $\Delta$ A vs B | $\Delta$ A vs C |
| AUT     | 18.93               | 21.99               | 20.13            | <b>3.06</b>     | 1.2             |

|     |       |       |       |             |      |
|-----|-------|-------|-------|-------------|------|
| DZA | 24.15 | 26.59 | 25.29 | 2.44        | 1.14 |
| MYS | 24.05 | 28.62 | 26.31 | <b>4.57</b> | 2.26 |
| PER | 27.63 | 30.04 | 28.86 | 2.41        | 1.23 |
| ROU | 25.04 | 27.47 | 26    | 2.43        | 0.96 |
| THA | 25.07 | 31.13 | 30.42 | <b>6.06</b> | 5.35 |

- Low risk mindset
- Experimental driven and maybe less purchasing power

Countries where Variant C (24/12/1) performed best

| Country | total_sessions<br>2 | Converted_sessions2 | Conversion_rate% |          |             |
|---------|---------------------|---------------------|------------------|----------|-------------|
| Country | CR_A (%)            | CR_B (%)            | CR_C (%)         | Δ A vs B | Δ A vs C    |
| ARG     | 35.08               | 36.69               | 37.2             | 1.61     | 2.12        |
| CZE     | 23.17               | 24.74               | 25.76            | 1.57     | 2.59        |
| HUN     | 25.34               | 22.82               | 25.8             | -2.52    | 0.46        |
| IRQ     | 15.17               | 13                  | 15.35            | -2.17    | 0.18        |
| KEN     | 32.55               | 29.49               | 34.09            | -3.06    | 1.54        |
| NGA     | 31.4                | 31.08               | 34.82            | -0.32    | <b>3.42</b> |
| SWE     | 23.49               | 23.74               | 26.32            | 0.25     | 2.83        |
| VNM     | 23.21               | 24.03               | 27.42            | 0.82     | <b>4.21</b> |

- Value driven, cost advantage
- High digital adoption, strong demand (censorship etc)

Performing statistical significance test to confirm practical meaningful uplift.

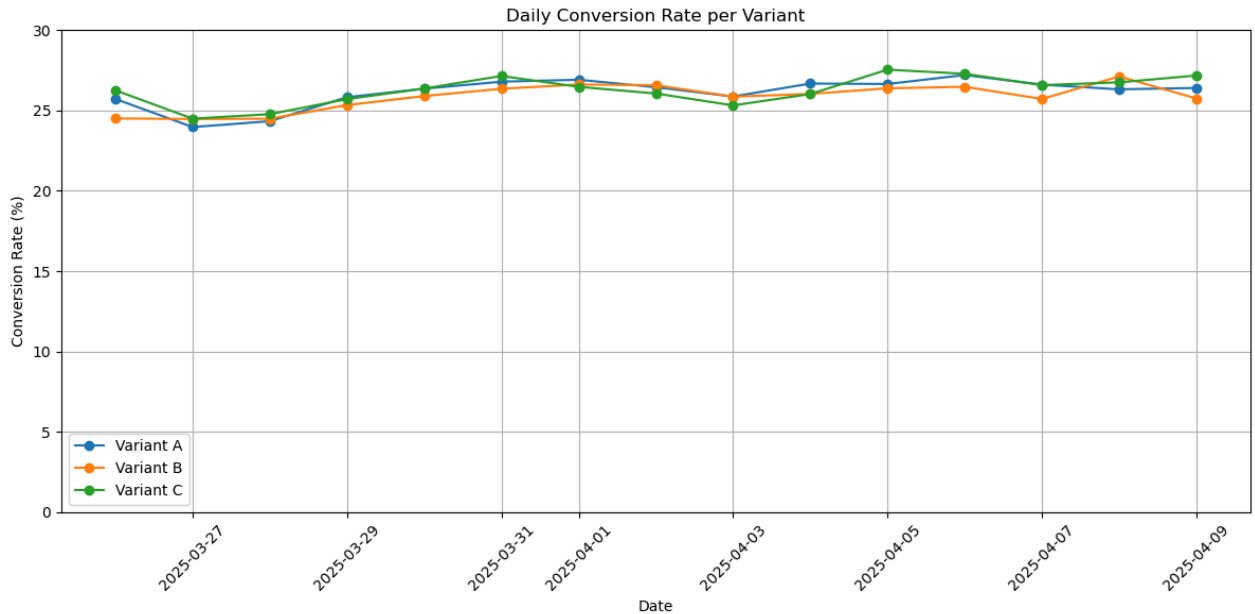
| Country | CR A (%) | CR B (%) | CR C (%) | Δ A→B        | p A vs B | Δ A→C | p A vs C | Recommendation       |
|---------|----------|----------|----------|--------------|----------|-------|----------|----------------------|
| THA     | 25.07    | 31.13    | 30.42    | <b>6.06</b>  | 0.0095   | 5.35  | 0.021    | Use Variant B        |
| MYS     | 24.05    | 28.62    | 26.31    | 4.57         | 0.0099   | 2.26  | 0.1917   | Use Variant B        |
| VNM     | 23.21    | 24.03    | 27.42    | <b>0.82</b>  | 0.5665   | 4.21  | 0.0032   | Use Variant C        |
| ARE     | 27.37    | 23.41    | 23.23    | <b>-3.96</b> | 0.0555   | -4.14 | 0.0418   | Stick with Variant A |

**Insights:**

- **Thailand and Malaysia** strongly favor **Variant B**, preferring to see **1-month plans first** — risk-averse, price-conscious trial-oriented behavior.
- **Vietnam** responds significantly better to **Variant C** — anchoring with a strong value proposition up front (24-month plan).
- **UAE** prefers **Variant A**, and statistically underperforms with the other two — suggesting the default order is best for user clarity and balance.

#### Time Series analysis

- Daily conversion Rate per variant



The chart shows that there are no major deviations in the daily conversion rates and conversion rates on weekends were slightly higher than on weekdays.

#### Bounce Rate per Variant

| Flag | Bounce Rate % |
|------|---------------|
| A    | 6.08          |
| B    | 6.24          |
| C    | 6.17          |

All variants have very similar bounce rates, with Variant B slightly higher. This suggests that plan order has minimal impact on immediate user drop-off behavior.

#### Average session durations

| Flag | Avg_session_duration_seconds |
|------|------------------------------|
| A    | 256.36                       |
| B    | 255.18                       |
| C    | 256.21                       |

Consistent across variants

Subscription to Sign Up Ratio (session based) (% of signups resulted in paid subscription)

| Variant | Signup Sessions | Paid Sessions | Subscription -to-Signup Ratio (%) |
|---------|-----------------|---------------|-----------------------------------|
| A       | 50,873          | 2,962         | 5.82%                             |
| B       | 50,258          | 2,916         | 5.80%                             |
| C       | 52,642          | 3,220         | 6.12%                             |

Variant C has the highest subscription to Sign up Ratio indicating better subscription efficiency among users who signup.

| Flag                            | A     | B     | C     |
|---------------------------------|-------|-------|-------|
| subscriptionsTotalByCycle_clean |       |       |       |
| 12:01.0                         | 20.28 | 22.4  | 19.99 |
| 01:01.0                         | 63.55 | 62.26 | 61.31 |
| 24:01.0                         | 16.17 | 15.29 | 15.41 |
| 03:01.0                         | NaN   | 0.05  | 3.29  |

The subscription plan of 1 month is the highest purchased plan across all variants. Followed by 12 month and 24 month plans